Bangwei Guo

E-mail · Google Scholar · Personal Website

Education

Rutgers, The State University of New Jersey, Piscataway, NJ, USA. Sep. 2023 – Present

Ph.D. Candidate in **Computer Science**.
Advised by Distinguished Prof. Dimitris N. Metaxas.

University of Science and Technology of China (USTC), Anhui, China. Sep. 2020 – Jun. 2023

Master in **Applied Statistics**.
Co-advised by Dr. Steven Xu and Prof. Hong Zhang.

University of Science and Technology of China (USTC), Anhui, China. Sep. 2016 – Jun. 2020

B.S. in **Mathematics and Applied Mathematics**.

Research Skills and Interests

- **Programming Languages:** Python, PyTorch, R, MATLAB, LaTeX, Git.
- **Research Areas:** Vision–Language Models, Visual Understanding, Medical Foundation Models, Universal and Interactive Segmentation, Multimodal Agents and Agentic Reasoning, Dataset Construction and Benchmarking.
- **Research Statement:** My research focuses on vision–language models and multimodal agents for structured visual understanding and reasoning, with an emphasis on scalable dataset construction and complex relational structure extraction. I also develop medical foundation models for vision–language understanding and universal segmentation across diverse modalities and tasks. Beyond these core areas, I contribute to a broad range of research projects on large language model (LLM) agents.

Experience

United Imaging Intelligence, Boston, MA, USA. May. 2026 – Aug. 2026
Research Intern. Mentored by Dr. Xiao Chen and Dr. Shanhui Sun.

- Developing a cardiac MRI vision-language model for fine-grained anatomical grounding and pathology understanding across multi-view, multi-sequence, and multi-phase CMR data.
- Building large-scale, quantitatively grounded CMR datasets by mining heterogeneous clinical reports and integrating reviewed segmentation masks, rule-based measurements, and literature-guided clinical thresholds.

NEC Laboratories America, Princeton, NJ, USA. Aug. 2025 – Present
Part-time Research Intern. Mentored by Dr. Xujiang Zhao.

- Developing VLM and agentic frameworks for structured relation and topology extraction from complex diagrams.
- Reinforcement learning fine-tuning of large language models for mathematical and combinatorial optimization.
- Building benchmark datasets and evaluation protocols for structured visual reasoning tasks.

Rutgers, The State University of New Jersey. Sep. 2023 – Present
Research Assistant at CBIM. Supervised by Distinguished Prof. Dimitris Metaxas.

- Developing medical foundation models for universal segmentation across organs, modalities, and interaction paradigms (semantic, in-context, and interactive segmentation).
- Designing scalable 2D/3D segmentation architectures that integrate prompt-based control and human-in-the-loop refinement.
- Exploring vision–language alignment and large vision–language models (VLMs) for medical image understanding and language-guided segmentation.
- 3D reconstruction of anatomical structures from multi-view and multi-modal medical images.

Teaching Assistant.

- Mathematical Foundations of Data Science; Introduction to Discrete Structures; Topics in Computers in Biomedicine; Introduction to AI.

University of Science and Technology of China. Sep. 2020 – Jun. 2023
Research Assistant.

- Developing image classification models for large-scale histopathology analysis, focusing on biomarker prediction and cancer progression assessment.
- Fine-tuning and training deep learning models across multiple cancer types and whole-slide image (WSI) datasets.
- Applying statistical modeling and simulation techniques to validate theoretical properties and model robustness.

Teaching Assistant.

- Statistical Calculation, *Fall 2021*.

Publications And Preprints

Vision-Language Models, LLMs, and AI Agents

- **Guo, B.**, Zhao, X., Chen, S., Liu, Y., Cheng, W., Zhu, X., Zhang, G., Metaxas, D. N., & Chen, H. (2026). Knossos and Ariadne: Benchmarking and Learning Complete Diagram Topology Extraction with Vision–Language Models. arXiv preprint arXiv:2610.04721. article.
- **Guo, B.**, Zhao, X., Liu, Y., Cheng, W., Chen, S., Li, D., Masaharu, M., Kuroda, T., Metaxas, D. N., & Chen, H. (2026). TopoAgent: A Structure-Aware Perception-to-Reasoning Framework for Diagram-to-Graph Topology Extraction with Large Vision-Language Models. arXiv preprint arXiv:2608.28701. article.
- Chen, X., Jin, M., Niu, J., Yin, Y., Zhao, J., **Guo, B.**, Metaxas, D. N., Wang, Z., Yue, Y., & Penn, G. (2026). All Circuits Lead to Rome: Rethinking Functional Anisotropy in Circuit and Sheaf Discovery for LLMs. (ICML 2026) article.
- Liu, Q., Jiao, Q., **Guo, B.**, Dai, W., Yang, C., Liu, J., Metaxas, D. N., Shi, Y., Jiang, Z., & Guo, M. (2026). Agents' Last Doubt: Knowing When to Act and When to Step Back in Tool-Using Agents. arXiv preprint. arXiv:2610.24194. article.
- Zhou, Y., Zhao, M., Wang, Z., Gu, D., **Guo, B**., Ye, R., Han, L., Jin, C. & Metaxas, D.N. (2025). M^3-Bench: Multi-Modal, Multi-Hop, Multi-Threaded Tool-Using MLLM Agent Benchmark. arXiv preprint arXiv:2511.17729. article.
- Gu, D., Gao, Y., Chatzoudis, G., Dong, Z., Zhang, G., **Guo, B.**, Zhou, Y., Zhou, M., & Metaxas, D. (2026). LUCID-SAE: Learning Unified Vision-Language Sparse Codes for Interpretable Concept Discovery. arXiv preprint arXiv:2602.07311. article
- Zhou, Y., Dong, Z., Wang, Z., Jin, C., Zhao, S., **Guo, B.**, ... & Metaxas, D. N. (2026). Evidence Over Plans: Online Trajectory Verification for Skill Distillation. arXiv preprint arXiv:2605.09192. article
- Cai, X., Hu, B., Jin, M., Zhang, Q., Ouyang, Y., **Guo, B.**, Guo, M., Yin, Y., Gao, L., Han, F., Yu, S., Liang, K., Zeng, Q., Niu, J., Yang, F., Cheng, W., Xiang, Z., McAuley, J., & Wang, Z. (2026). RealControlBench: A Benchmark for Interpretability-Informed Fine-Grained Control of LLMs and Agents. arXiv preprint. arXiv:2610.27295. article.
- Cai, X., Guo, M., Liu, J., Han, J., **Guo, B.**, Long, Y., ... & Li, R. (2026). OpenPM: Auditable Point-in-Time Evaluation for LLM Portfolio-Management Agents. (SIGKDD Explorations). article

Medical AI and Computer Vision

- **Guo, B.**, Chen, X., Mailhe, B., Yao, J., Wang, Y., Mukherjee, A., Liu, Y., Zhang, Z., Yu, H., Chen, T., & Sun, S. (2026). Learning Where to Look: Anatomical Grounding and Guided Attention for Cardiac MRI Vision-Language Models. arXiv preprint arXiv:2609.39899. article.
- **Guo, B.**, Gao, Y., Ye, M., Gu, D., Zhou, Y., Zhang, G., Axel, L., & Metaxas, D. (2026). UniPro: Unified Multi-Mode Medical Image Segmentation from 2D Images to 3D Volumes via Propagation. (NeurIPS 2026) article.
- **Guo, B.**, Gao, Y., Ye, M., Gu, D., Zhou, Y., Axel, L., & Metaxas, D. (2025). K-Prism: A Knowledge-Guided and Prompt Integrated Universal Medical Image Segmentation Model. (ICLR 2026) article

- **Guo, B.**, Ye, M., Gao, Y., Xin, B., Axel, L., & Metaxas, D. (2025). VerSe: Integrating Multiple Queries as Prompts for Versatile Cardiac MRI Segmentation. (IPMI 2025). article
- **Guo, B.**, Li, X., Yang, M., Jonnagaddala, J., Zhang, H., & Xu, X. S. (2023). Predicting microsatellite instability and key biomarkers in colorectal cancer from H&E stained images: achieving state of the art predictive performance with fewer data using swin transformer. The Journal of Pathology: Clinical Research, 9(3), 223-235. article.
- **Guo, B.**, Li, X., Yang, M., Zhang, H., & Xu, X. S. (2023). A robust and lightweight deep attention multiple instance learning algorithm for predicting genetic alterations. Computerized Medical Imaging and Graphics, 105, 102189. article
- Zhao, A., Ye, M., Gu, D., **Guo, B.**, Myers, M., Lock, H., Eckert, G., Dutra, V., Metaxas, D., & Turkkahraman, H. (2025). Reconstruction of 3D Models of the Mandible from 2D Lateral Cephalometric Radiographs by Deep Learning. Research Square. article
- Axel, L., Kanski, M., Jhaveri, A., Ye, M., **Guo, B.**, He, X., & Metaxas, D. (2025). Analysis of Cardiac Dynamic Global Function. In International Conference on Functional Imaging and Modeling of the Heart (pp. 197-206). article
- Ye, M., Xin, B., **Guo, B.**, Axel, L., & Metaxas, D. (2024). Learning Volumetric Neural Deformable Models to Recover 3D Regional Heart Wall Motion from Multi-Planar Tagged MRI. arXiv preprint arXiv:2411.15233. article.
- Liu, A., Li, X., Wu, H., **Guo, B.**, Jonnagaddala, J., Zhang, H., & Xu, S. (2023). Prognostic Significance of Tumor-Infiltrating Lymphocytes Determined Using LinkNet on Colorectal Cancer Pathology Images. JCO Precision Oncology, 7, e2200522. article
- Cen, M., Li, X., **Guo, B.**, Jonnagaddala, J., Zhang, H., & Xu, X. S. (2023). A Novel and Efficient Digital Pathology Classifier for Predicting Cancer Biomarkers Using Sequencer Architecture. The American Journal of Pathology, 193(12), 2122-2132. article
- Yuan, M., Ding, H., **Guo, B.**, Yang, M., Yang, Y., & Xu, X. S. (2024). Image-Based Subtype Classification for Glioblastoma Using Deep Learning: Prognostic Significance and Biologic Relevance. JCO Clinical Cancer Informatics, 8, e2300154. article